

LIONAEL SURYA DWITAMA

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PROFESSIONAL SUMMARY

Data Science & Analytics student at Seneca Polytechnic (Cumulative GPA 3.9/4.0, President's Honour List) with hands-on experience building and validating statistical and machine-learning models on real-world datasets — from a walk-forward-validated quantitative trading system on the IDX Kompas100 universe to an explainable XGBoost flood-risk model. Proficient in Python and SQL for data manipulation, model development, and turning technical analysis into client-ready outputs; looking to bring that same analytical rigor to Oliver Wyman's Data and Analytics practice this summer.

EDUCATION

Seneca Polytechnic — Newnham Campus, Toronto, ON

2024 – Present (Expected Apr. 2028)

Honours Bachelor of Data Science and Analytics

- Cumulative GPA: 3.9/4.0; President's Honour List (Winter 2025, Winter 2026)
- Relevant coursework: Statistical Methods for Data Science, Data Mining, Simulation & Modelling, Business Intelligence, Data Visualization, Advanced Database Design

TECHNICAL SKILLS

Languages & Querying: Python, SQL, JavaScript

Statistical & ML Modeling: XGBoost, SHAP, Optuna hyperparameter tuning, walk-forward validation, statistical analysis

Data & Visualization: Power BI, Data Visualization, Microsoft Excel, Leaflet.js dashboards

Databases: PostgreSQL, MySQL, Supabase

Tools: Git/GitHub, VS Code, GitHub Actions, Microsoft Teams/Outlook, Google Workspace

PROJECTS

Kompas100 Quantitative Trading System — ISTC 2026 Competition

2026

- Built a walk-forward-validated quantitative trading system for the IDX Kompas100 stock universe, competing in a 14-trading-day contest scored on both P&L performance and a judged pitch presentation.
- Benchmarked candidate ML trading signals against a naive 20-day momentum baseline using a dedicated walk-forward evaluation engine, holding models to that baseline rather than reporting in-sample results alone.
- Designed a position concentration and sizing policy on the same walk-forward engine, reframing the objective from expected-return maximization to a top-7-finish optimization problem.
- Partnered with a co-developer to divide model development and portfolio-construction work against a fixed model-freeze deadline.

IDX Stock Scanner Agent — github.com/suryalionael/idx-stock-scanner-agent

- Designed a three-layer analytics pipeline — rule-based filtering, XGBoost-based ranking, and LLM-generated explanations via the Anthropic Claude API — to scan the Indonesian stock market (IDX) across Swing, Scalping, and Long Term modes.
- Ran a full technical audit of the tool's model and data layers; fixed a trading-calendar gap with a new IDX holiday-aware module and replaced deterministic mock broker data with a live Index Alpha API integration, including quota-safe batch fetching and cached Parquet storage.
- Used the audit results to grade each mode's reliability rather than presenting all outputs with equal confidence, prioritizing remaining model-validation work accordingly.
- Stack: Python, yfinance, XGBoost, Parquet, GitHub Actions.

FloodCast Jakarta — Dicoding AI Impact Challenge 2026

2026

- Built a multi-horizon XGBoost ensemble to predict urban flood risk in Jakarta, using SHAP for prediction-level explainability and Optuna for hyperparameter tuning.
- Developed an interactive Leaflet.js dashboard to communicate flood-risk predictions geographically to non-technical stakeholders.

EXPERIENCE

Software Engineering Intern — CV. Veda Sakti Dharma

Jun 2025 – Aug 2025

Jakarta, Indonesia

- Developed and tested a Python-based payroll tax (PPH 21) application to automate employee tax calculations.
- Debugged issues and validated payroll data to improve system reliability and reporting accuracy, collaborating with the team to implement new features.

Data Entry and Web Development — QuickRn Canada

Jul 2025 – Present

- Investigated and resolved website and online-course access issues through technical troubleshooting and cross-team collaboration.
- Responded to customer inquiries via email, documenting technical issues and communicating resolutions professionally.